

An-Sheng (Anson) LU

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EDUCATION

University of California, Riverside, CA, USA | *Expected Spring 2029* **2025 – Present**

Major in Electrical Engineering at Department of Electrical and Computer Engineering | Cumulative GPA: 3.87.

Relevant Coursework: Data Structures & Algorithms (C++), Logic Design, Physics, Multivariable Calculus.

Awards and Honors: UCR Achievement Scholarship (2025–29), Dean's Honors List.

Victoria Academy, Yunlin, Taiwan

2022 – 2025

IB Bilingual Diploma, July 6, 2025 | GPA: 3.94 (IBDP) / 3.86 (Cumulative).

– International Baccalaureate Diploma Programme (IBDP) HL subjects: Mathematics AA (7), Physics (6), Chemistry (6).

Awards: Chairman's/Mayor's Award, Dr. Cho Rui-Chiao Scholarship, Excellence Award in Mathematics, School Scholarships (5x recipient, 2022–2024).

EXPERIENCES

Research Assistant, Robotics and Medical Systems (RaMS) Lab, University of California, Riverside **Oct. 2025 – Present**

- Independently developed a CV/DL pipeline using AISFormer (deep-learning instance segmentation model) in collaboration with the UCR Entomology Dept., automating biomass estimation; awarded UCR Research Mini-Grant to fund project development.
- Architected automated workflows featuring normal distribution filtering and post-processing analysis to translate image data into high-accuracy phenotypic measurements for life-cycle analysis.

Intern, Electrical Subsystems, Highlander Racing (FSAE), University of California, Riverside

Apr. 2026 – Present

- Contributed to the design of an industrial automation PCB in Altium Designer, from schematic capture of analog sensing stages to routing power stages.

Research Assistant, Biomechanics and Biomaterials Design Laboratory, University of California, Riverside **May 2026 – Aug. 2026**

- Resolved signal and flow instability in a Right Heart Simulator (Physical Flow Testbed) by restructuring circuit schematics, conducting hydrodynamic diagnostics, and designing the wire layout for an Arduino Uno R3 shield.
- Designed custom SolidWorks mechanical enclosures, including a redesigned atrial reservoir chamber cover and circuit box housing, to improve structural integrity and fluid containment.

Inventor, Taiwan Intellectual Property Office, Taiwan

June 2024 – Aug. 2025

- Patent: *Baton Device with Directional Indication* (Invention Patent No. I894025). [Taiwan Patent Search System](#).
- Independently designed a signaling device with custom hardware logic using an ESP32 and MPU-6050 (IMU) to process real-time indication; managed conceptual design and circuit prototyping phases through to formal patent approval.

Intern, Robotics and Mechatronics Center, German Aerospace Center (DLR), Oberpfaffenhofen, Germany

Aug. 1 – 30, 2024

- Integrated a custom robotic system using an OpenManipulator-X arm; collaborated with advisors on system design and implemented 3D-printed structural interfaces and motion control.
- Modeled mass-spring-damper systems in MATLAB/Simulink to analyze resonance and power dissipation; designed a critically damped feedback control system to optimize stability.
- Designed and assembled a 3x3x3 LED matrix using an Arduino Micro; implemented a 4-mode firmware system featuring real-time pace control via ADC and a variable resistor.

Robotics Club Member, National Yunlin Univ. of Sci. & Tech. (NYUST), Yunlin, Taiwan

Sep. 2023 – June 2024

- Collaborated with club members to program an omnidirectional mobile robot with a three-Mecanum wheel chassis; executed sensor fusion using Ultrasonic and Color Detection sensors for obstacle avoidance and environmental interaction via LabVIEW.
- Implemented basic ROS interfacing for system communication.

HONORS & AWARDS

- 1st Place Winner, Robotics Competition, Yunlin, Taiwan, Dec. 25, 2023.
- 1st Place Winner, Industrial Robot Competition, Kaohsiung, Taiwan, Mar. 9, 2024.
- Final Shortlisted Prize, Electrical Engineering Category, 2024 All Ring Competition, Tainan, Taiwan, Oct. 24, 2024

SKILLS

- Language:** Mandarin (Native), English (Fluent), Taiwanese/Min Nan Chinese (Conversational), German (Beginning)
- Programming:** C/C++, Python (PyTorch, OpenCV), LabVIEW, MATLAB/Simulink, ROS (Basic).
- Software & Hardware Skills:** Tinkercad, Creo, Arduino (IDE), Visual Studio/VS Code, Altium Designer, Solidworks.
- Interests:** Basketball (Clippers), Formula 1 (Red Bull Racing), Strategic Games (Chess/Xiangqi), Robotics, Electronics.